

Load Interconnection Request System Impact Study Report

For Queue Project Q21-1702

Roanoke Rapids 138 kV
June 27, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1702. The Interconnection Customer proposes a 302.3 MW large load (data center) facility interconnecting at the Roanoke Rapids 138 kV substation in the North Carolina area. The request was received on March 26, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 549 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

No thermal overloads attributable to this project were identified. No network upgrades are required. Estimated network upgrade cost: \$0

3. Network Impacts

3.1 Load Deliverability Analysis

No overloads identified.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

None. Total network upgrade cost allocated to this project: \$0.

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.